

Rahil Parikh

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Robotics researcher building learned controllers and perception-conditioned policies that bridge real-time embedded systems and modern reinforcement learning. Trained PPO quadruped locomotion policies in NVIDIA Isaac Lab on AnymalC with terrain curricula and contact-aware reward shaping; ships production C++/Qt real-time control software (47,000+ LOC) running closed-loop on Jetson hardware in customer labs. Incoming MSE Robotics student at Penn GRASP (Fall 2026), targeting research at the intersection of visuomotor RL, sim-to-real transfer, and learned-vs-classical controller analysis.

EDUCATION

University of Pennsylvania, Philadelphia, PA

Incoming Fall 2026

M.S.E. in Robotics (GRASP Lab), School of Engineering and Applied Science

Worcester Polytechnic Institute, Worcester, MA

August 2019 – May 2023

B.S. in Computer Science and Robotics Engineering, GPA: 3.8/4.0

- **Coursework:** Robot Dynamics, Motion Planning, Machine Learning, Computer Vision, Linear Systems, Feedback Control, Embedded Systems, Algorithm Design, AI Fundamentals

TECHNICAL SKILLS

Robot Learning & Simulation: NVIDIA Isaac Lab, Isaac Gym, MuJoCo, Gazebo, RSL-RL, Stable-Baselines3, PPO, reward shaping, terrain curricula, sim-to-real transfer, domain randomization, imitation learning, RunPod / A100 training

Control & Estimation: Real-time control systems, MPC formulation, PID, Kalman filtering, state estimation, contact dynamics, trajectory optimization, inverse kinematics, ROS/ROS2, RViz, Lie groups (SO(3)/SE(3)) for VIO/SLAM

Perception & Computer Vision: PyTorch, OpenCV, MediaPipe, point cloud processing, LiDAR sensor fusion, feature extraction, object detection, camera calibration, vision-conditioned policy learning

Programming: C++ (production, multithreaded), Python, MATLAB, CUDA, Java, SQL

Embedded & Systems: NVIDIA Jetson (Nano, Orin NX), Embedded Linux, Yocto, Qt/QML, ZeroMQ, FFTW, FPGA hardware interlocks, Docker, Git

Currently Studying: Drake (contact-implicit trajectory optimization), Vision-Language-Action models, diffusion policies, micro Lie theory for state estimation

RESEARCH EXPERIENCE

Quadruped Locomotion via Deep RL | *NVIDIA Isaac Lab, PyTorch, RSL-RL, PPO, RunPod A100* November 2025 – Present

- Trained four PPO locomotion policies on AnymalC in NVIDIA Isaac Lab with terrain curricula, achieving stable trotting gaits across all four (duty cycle 0.33–0.62, diagonal foot correlation -0.56 to -0.90) verified via gait analysis on held-out terrain
- Designed and ran a contact-aware reward ablation study isolating per-term contribution; identified `terrain_clearance` as most valuable (-20.2% task return when removed) and `foot_slip` as most constraining (removal increased reward $+32.6\%$ but halved walking distance), quantifying the reward-vs-behavior gap
- Diagnosed and fixed a silent rsl-rl v5 checkpoint-loading bug where actor weights saved as `mlp.0.weight` loaded into a `net.0.weight` MLP, masking near-zero locomotion behind a standing-bonus reward; resolved via `OnPolicyRunner.load() + get_inference_policy()` in a shared `runner_utils.py`
- Currently extending the pipeline toward a vision-conditioned policy reproducing Agarwal et al. (CoRL 2022) “Legged Locomotion in Challenging Terrains using Egocentric Vision”

Contact-Implicit Trajectory Optimization Reproduction | *Drake, C++, Python* Planned May – August 2026

- Reproducing the C3 (Consensus Complementarity Control) algorithm from Penn DAIR Lab in Drake, targeting MPC-style contact-rich manipulation as preparation for GRASP coursework and faculty outreach to Prof. Posa

PROFESSIONAL EXPERIENCE

Control Systems Software Engineer | *Bridge12 Technologies, Natick, MA*

September 2023 – Present

- Author and maintainer of a 47,000+ line Qt/C++ real-time control system orchestrating 8+ heterogeneous instruments via a multi-device finite state machine, deployed in production at customer research facilities
- Cut closed-loop network latency 70% (5.0s $\rightarrow$ 1.5s) by replacing synchronous polling with a ZeroMQ asynchronous publisher and Kalman-filtered distributed state estimation, enabling stable feedback at higher control bandwidth
- Built a real-time search-and-track algorithm processing 100,000+ data points in under 75 ms using spatial hashing and k-d trees, sustaining hard real-time guarantees for dynamic system tracking

- Engineered Yocto-based custom Linux distributions for NVIDIA Jetson Nano and Orin NX, tuning kernel and boot configuration for deterministic latency on embedded control hardware
- Owned on-site deployment, debugging, and operator training at 5+ customer research sites (incl. Columbia University Havemeyer Hall), translating ambiguous physics requirements into shippable closed-loop systems

Software Engineering Intern | *Ekotrope, Boston, MA*

May 2022 – October 2022

- Improved MariaDB transaction throughput 20% via query and schema redesign on a physics-based building energy modeling backend; cut modeling defect rate 18% through JUnit validation suites against analytical baselines

SELECTED PROJECTS

Bipedal Humanoid Robot | *Python, C++, ROS, Gazebo, RViz*

August 2022 – August 2024

- Led 12-person interdisciplinary team building a bipedal humanoid for dynamic walking; secured \$15,000 in prototype funding through a faculty advisory pitch cycle
- Implemented inverse-kinematics gait generation with ZMP-based stability feedback, achieving sub-50 ms reactive balancing response on hardware

Autonomous Mobile Robot Navigation | *C++, ROS, LiDAR, AMCL, GMapping* October 2021 – December 2021

- Built full autonomy stack (SLAM, particle-filter localization, A* / move_base planning) with Kalman-filtered LiDAR-odometry fusion; demonstrated collision-free indoor navigation as lead software developer

Robotic Arm Vision & Motion Planning | *C++, MATLAB, OpenCV*

August 2021 – October 2021

- Built a forward/inverse-kinematics pipeline with an OpenCV pose-estimation front-end driving a 6-DOF manipulator to fully autonomous color-sorted pick-and-place

Hand-Gesture Robotic Control | *Python, MediaPipe, OpenCV*

2024

- Built a real-time vision pipeline mapping MediaPipe hand-landmark estimates to control signals, exploring perception-as-interface for teleoperation

SELF-STUDY & SCHOLARSHIP

- **Micro Lie Theory for Robotics** (Solà et al.) – working knowledge of $SO(3)/SE(3)$ Lie groups, exponential/log maps, adjoint, right/left Jacobians, and EKF/SLAM/VIO applications
- **Google DeepMind “Measuring Progress Toward AGI” Kaggle Hackathon** – benchmarks in Metacognition and Executive Functions tracks (Hayling Sentence Completion, IOED, calibration asymmetry)
- **Reading list, May–Aug 2026:** Agarwal et al. (CoRL 2022) vision-conditioned legged locomotion; Kunapuli et al. (2025) classical vs. learned quadrotor controllers; Hu et al. (NeurIPS 2025) interactive perception RL; Posa et al. on contact-implicit MPC

HONORS & LEADERSHIP

- **WPI Presidential Scholarship** – merit award for incoming students; **WPI Dean’s List**
- **Upsilon Pi Epsilon (UPE)** – International Honor Society for Computing
- **WPI SWEET Fellowship** – selected for research dedication and technical communication
- **Runner-Up Alliance, FIRST Robotics Competition India (2018)**